VISUAL JIGSAW POST-TRAINING IMPROVES MLLMS

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ABSTRACT

Current reinforcement learning (RL) based post-training for multimodal large language models (MLLMs) primarily focuses on text-centric reasoning tasks, using visual inputs as sparse evidence for chain-of-thought processes. This approach overlooks the intrinsic visual perception capabilities that require fine-grained spatial, temporal, and depth understanding. We introduce **Visual Jigsaw**, a self-supervised post-training framework that enhances visual perception in MLLMs through verifiable permutation prediction tasks. Our method partitions visual inputs—images into spatial patches, videos into temporal clips, and 3D scenes into depth-ordered point sets—then challenges models to predict correct orderings using only text outputs. Trained with Group Relative Policy Optimization (GRPO), Visual Jigsaw employs a graded reward system combining exact-match accuracy, discounted partial correctness, and format constraints to prevent reward hacking. Extensive experiments on Qwen2.5-VL-7B demonstrate substantial improvements across diverse benchmarks: +6.00 on MMVP, +6.06 on MMStar fine-grained tasks, +5.23 on AoTBench, +17.11 on DA-2K, and +15.34 on SAT-Real. Crucially, Visual Jigsaw strengthens perception without degrading reasoning capabilities, making it complementary to existing reasoning-focused RL methods. Our approach requires no architectural modifications or additional generative components, offering a simple yet effective recipe for enhancing visual understanding in MLLMs.

1 INTRODUCTION

Multimodal large language models (MLLMs) have demonstrated remarkable capabilities in understanding and reasoning about visual content, yet a significant gap remains between their performance on text-centric reasoning tasks and fine-grained visual perception benchmarks. Current reinforcement learning (RL) post-training approaches for MLLMs predominantly focus on enhancing reasoning abilities through chain-of-thought processes, where visual inputs serve merely as sparse contextual evidence Shao et al. (2024); Chen et al. (2025). This text-centric paradigm, while effective for complex reasoning tasks, systematically under-optimizes the intrinsic visual perception capabilities that are fundamental to multimodal understanding.

The challenge lies in designing RL objectives that directly target visual perception while remaining compatible with text-only output generation—a constraint that rules out generative visual tasks or dense prediction objectives. Traditional visual perception tasks often require specialized architectures, additional modalities, or dense outputs, making them difficult to integrate into existing MLLM training pipelines.

We address this challenge by introducing **Visual Jigsaw**, a novel self-supervised post-training framework that recasts visual understanding as a lightweight, verifiable ordering task solvable through text outputs. Our key insight is to partition visual inputs into meaningful components—spatial patches for images, temporal clips for videos, and depth-ordered point sets for 3D scenes—then shuffle these elements and task the model with recovering the correct permutation.

Visual Jigsaw offers several distinctive advantages: (1) *Self-supervised and verifiable*: The task generates its own supervision through shuffling, with objectively verifiable correct answers. (2) *Text-output compatible*: Models predict permutations as text sequences, requiring no architectural modifications. (3) *Modality-agnostic*: The framework naturally extends across images, videos, and

3D data with consistent formulation. (4) *Perception-centric*: Unlike reasoning tasks, jigsaw completion directly exercises spatial, temporal, and depth understanding.

To train Visual Jigsaw effectively, we employ Group Relative Policy Optimization (GRPO) Shao et al. (2024) with a carefully designed graded reward function. Our reward combines exact-match accuracy for perfect predictions with discounted partial correctness for near-misses, while enforcing strict format constraints to prevent reward hacking through invalid permutations.

We conduct extensive experiments using Qwen2.5-VL-7B Bai et al. (2025) as our base model, demonstrating consistent improvements across diverse visual understanding benchmarks. Visual Jigsaw achieves substantial gains on fine-grained perception tasks (+6.00 on MMVP, +6.06 on MM-Star), temporal understanding (+5.23 on AoTBench), and 3D spatial reasoning (+17.11 on DA-2K, +15.34 on SAT-Real). Importantly, these improvements come without degrading performance on reasoning-intensive tasks, indicating that Visual Jigsaw complements rather than conflicts with existing reasoning-focused RL methods.

Our contributions are threefold: (1) We introduce Visual Jigsaw, a novel self-supervised framework that enhances visual perception in MLLMs through permutation prediction tasks. (2) We demonstrate that RL-based training significantly outperforms supervised fine-tuning for this task, with careful reward design proving crucial for stable learning. (3) We provide extensive empirical validation showing broad improvements in visual understanding while preserving reasoning capabilities, establishing Visual Jigsaw as an effective complement to reasoning-focused post-training approaches.

2 Related Works

Multimodal Large Language Models. Recent advances in multimodal large language models have achieved impressive performance across diverse vision-language tasks Wang et al. (2024); Bai et al. (2025). These models typically integrate pre-trained vision encoders with large language models, enabling them to process and reason about visual content through natural language interfaces. However, despite their success on high-level reasoning tasks, MLLMs often struggle with fine-grained visual perception that requires detailed spatial, temporal, or depth understanding Tong et al. (2024).

Reinforcement Learning for Language Models. Reinforcement learning has become a cornerstone of post-training for large language models, particularly through methods like PPO and its variants Shao et al. (2024). Recent work has extended RL training to multimodal settings, primarily focusing on improving reasoning capabilities through chain-of-thought processes. However, these approaches typically treat visual inputs as contextual evidence rather than targets for direct perception enhancement, leaving substantial room for improvement in visual understanding capabilities.

Visual Perception and Understanding. Visual perception encompasses multiple dimensions including spatial understanding, temporal dynamics, and depth perception. Traditional computer vision approaches address these through specialized architectures and dense prediction tasks. In the context of MLLMs, visual perception benchmarks like MMVP Tong et al. (2024) have revealed significant gaps between model performance on reasoning tasks versus fine-grained perception tasks. Our work specifically targets these perception capabilities through a unified framework that addresses spatial, temporal, and 3D understanding.

Self-supervised Learning for Vision. Self-supervised learning has proven highly effective for visual representation learning, with jigsaw puzzle solving being a classic pretext task that requires understanding spatial relationships Noroozi & Favaro (2016). However, traditional jigsaw approaches are designed for dense visual representations and cannot be directly applied to text-generating MLLMs. Our Visual Jigsaw framework adapts the core insight of spatial relationship learning to the text-generation paradigm while extending it to temporal and depth dimensions.

Graded Reward Systems. Effective RL training often requires carefully designed reward functions that balance different objectives while preventing reward hacking. Recent work in RL for language models has emphasized the importance of graded rewards that capture partial progress toward

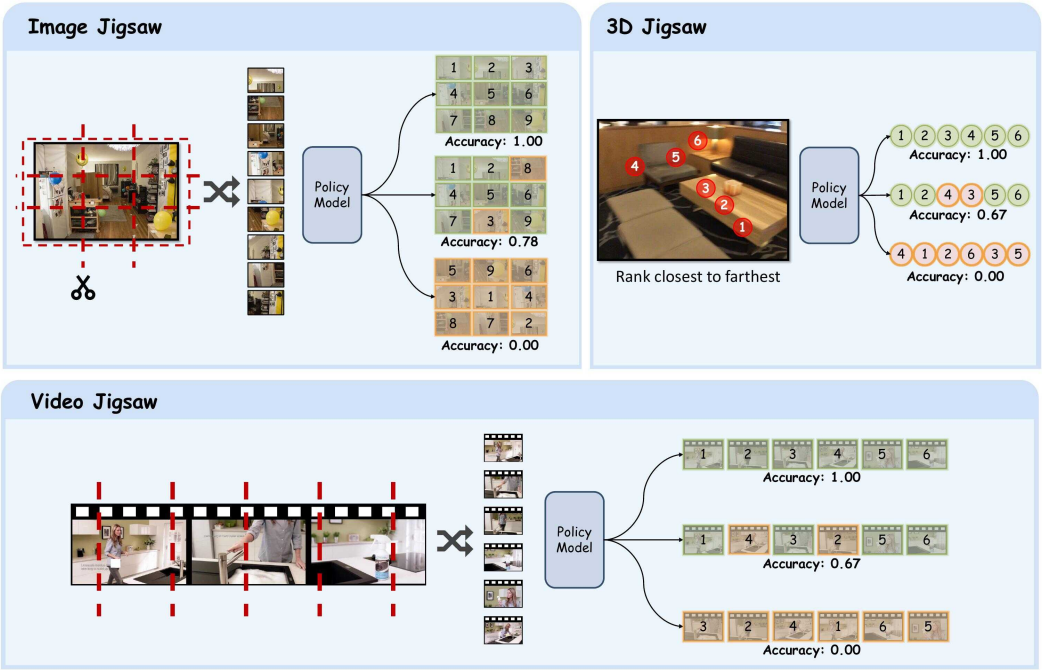

Figure 1: **Illustration of the Visual Jigsaw tasks.** Our method partitions visual inputs into meaningful components (spatial patches for images, temporal clips for videos, depth-ordered sets for 3D) and challenges models to predict correct orderings through text outputs. The figure shows examples for all three modalities with their corresponding accuracy-based rewards.

desired behaviors Shao et al. (2024). Our approach builds on these insights by designing a reward system that combines exact-match accuracy with discounted partial correctness, while maintaining strict format constraints to ensure valid outputs.

3 METHOD

Visual Jigsaw transforms visual understanding into a verifiable permutation prediction task that can be solved through text generation. We partition visual inputs into meaningful components, shuffle them, and task the model with predicting the correct ordering. This section details our approach across three modalities: images, videos, and 3D scenes.

3.1 VISUAL JIGSAW TASK FORMULATION

For any visual input V, we define a partitioning function $\mathcal{P}$ that divides V into an ordered sequence of components $C = [c_1, c_2, \ldots, c_k]$. We then apply a random permutation π to obtain shuffled components $C_\pi = [c_{\pi(1)}, c_{\pi(2)}, \ldots, c_{\pi(k)}]$. The task is to predict the permutation π^{-1} that restores the original ordering.

Given the shuffled visual input and a text prompt, the model generates a response containing the predicted ordering as a sequence of indices. The correct answer is the permutation that maps the shuffled order back to the canonical order.

3.2 MODALITY-SPECIFIC INSTANTIATIONS

Image Jigsaw. For images, we implement spatial partitioning by dividing the input image into a regular $m \times n$ grid (typically 3×3). Using our slicing function detailed in the code implementation, we ensure lossless reconstruction by assigning extra pixels to boundary tiles. The canonical order

follows row-major traversal (reading order), providing an intuitive spatial prior. The model views the shuffled patches and must predict the ordering that reconstructs the original spatial layout.

Video Jigsaw. For videos, we extract k non-overlapping temporal clips using uniform sampling with boundary trimming to avoid edge effects. Each clip maintains its internal temporal structure while being positioned out of chronological order. The canonical order is the original temporal sequence. We optionally apply spatial partitioning within each clip, creating a hierarchical jigsaw where both temporal ordering and spatial arrangement must be recovered.

3D Jigsaw. For 3D scenes represented as RGB-D images, we sample point sets from the depth map and rank them by distance from the camera. This creates a natural ordering based on depth relationships. The model must understand 3D spatial structure to correctly rank points from nearest to farthest, exercising depth perception and 3D spatial reasoning.

3.3 Training with GRPO and Graded Rewards

We train Visual Jigsaw using Group Relative Policy Optimization (GRPO) Shao et al. (2024), a variant of PPO that enhances memory efficiency while maintaining training stability. The key to effective learning lies in our carefully designed reward function that balances accuracy with partial credit.

Graded Reward Design. Our reward function $R(\pi_{pred}, \pi_{true})$ combines three components:

$$R(\pi_{pred}, \pi_{true}) = \alpha \cdot \mathbb{I}[\pi_{pred} = \pi_{true}] \tag{1}$$

$$+ \beta \cdot \gamma^{d(\pi_{pred}, \pi_{true})} \cdot \mathbb{I}[\text{valid}(\pi_{pred})] \tag{2}$$

$$+ \lambda \cdot \mathbb{I}[\text{format}(\pi_{pred})] \tag{3}$$

where $\mathbb{I}[\pi_{pred} = \pi_{true}]$ is the exact-match indicator, $d(\pi_{pred}, \pi_{true})$ measures the edit distance between predictions, $\gamma = 0.2$ is the discount factor for partial credit, and the format term ensures valid permutation structure.

This design provides strong incentives for correct answers while offering graduated rewards for partially correct predictions. The format constraint prevents reward hacking through invalid outputs like repeated indices or non-permutations.

Training Stability. The verifiable nature of jigsaw tasks eliminates ambiguity in reward assignment, leading to stable RL training. Unlike open-ended generation tasks where reward models may be inconsistent, our ground-truth permutations provide clear optimization signals. This stability allows aggressive learning rates and efficient convergence.

3.4 Implementation Details

We implement Visual Jigsaw as a plug-and-play post-training method requiring no architectural modifications to existing MLLMs. The training pipeline processes visual inputs through modality-specific partitioning functions, constructs prompts with shuffled components, and optimizes the model to generate correct permutation predictions.

For computational efficiency, we pre-compute jigsaw puzzles offline and cache them during training. This approach allows us to control puzzle difficulty and ensure balanced coverage across different visual patterns and structures.

4 Experiments

We conduct comprehensive experiments to evaluate Visual Jigsaw across diverse visual understanding benchmarks. Our evaluation covers three key questions: (1) Does Visual Jigsaw improve visual perception across modalities? (2) How does RL training compare to supervised fine-tuning? (3) Can Visual Jigsaw enhance reasoning-oriented models without harming their capabilities?

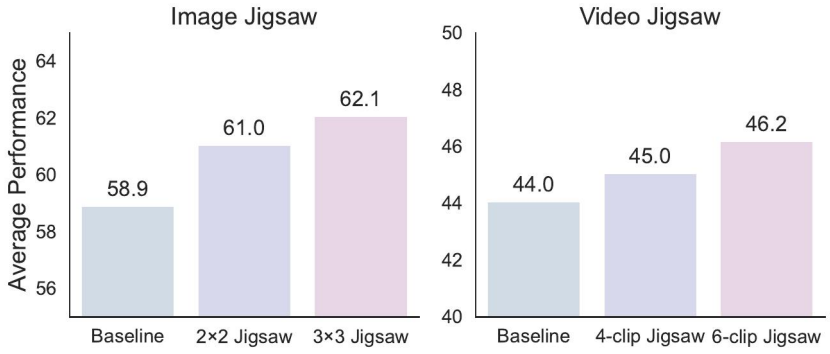

Figure 2: **Performance with different jigsaw difficulties.** Results show that increased difficulty (more patches/clips) leads to better performance, suggesting that harder puzzles provide richer training signal for visual perception.

4.1 Experimental Setup

Base Model. We use Qwen2.5-VL-7B Bai et al. (2025) as our base model, a state-of-the-art multimodal LLM with strong performance across vision-language tasks. This choice allows us to isolate the impact of Visual Jigsaw post-training from architectural advantages.

Training Configuration. We train with GRPO using a learning rate of 1×10^{-6}, batch size of 256, and gradient accumulation to maintain training stability. The reward function uses $\alpha = 1.0$ for exact matches, $\beta = 0.5$ for partial credit with $\gamma = 0.2$ discount, and $\lambda = 0.1$ for format compliance.

Jigsaw Configurations. For images, we use 3×3 spatial grids unless specified otherwise. For videos, we extract 4-6 temporal clips with optional spatial partitioning. For 3D scenes, we sample 6-8 depth-ordered point sets from RGB-D inputs.

4.2 Image Understanding Results

Table 1: **Evaluation results on image benchmarks.** Image Jigsaw achieves consistent improvements across fine-grained perception, spatial understanding, and compositional understanding tasks.

	Fine-grained Perception & Understanding								Spatial Und (Mono)			Compositional Und	
Model	**MMVP**	**MMStar (fine-grained)**	**MMBench**	**HR-Bench-8K**	**V***	**MME-RealWorld**	**LISA-Grounding**	**OVD-Eval**	**VSR**	**OmniSpatial**	**DA-2K**	**Winoground**	**SugarCrepe++**
	test	fine	en_dev	test	test	lite	test	test	test	test	val	g-acc	test
ThinkLite-VL	55.33	59.95	84.19	68.12	76.96	46.17	73.70	35.78	78.09	42.60	58.46	35.25	61.49
VL-Cogito	55.33	56.64	82.98	69.62	79.58	47.63	72.26	35.78	79.82	44.29	56.43	38.25	63.59
LLaVA-Critic-R1	53.33	57.80	83.16	67.50	78.01	45.18	68.52	35.28	78.50	42.73	53.82	34.75	61.93
Qwen2.5-VL-7B	54.66	59.75	83.33	67.38	76.96	43.41	71.89	35.07	77.68	42.66	54.45	37.00	61.59
Image Jigsaw (SFT)	56.00	60.94	83.67	69.75	80.10	43.88	66.59	34.35	80.68	43.55	61.46	38.75	62.03
Image Jigsaw	60.66	65.81	84.45	71.13	80.63	45.96	74.54	36.49	80.36	44.49	60.35	39.00	63.02
(Gain)	+6.00	+6.06	+1.12	+3.75	+3.66	+2.55	+2.65	+1.42	+2.68	+1.83	+5.90	+2.00	+1.43

Table 1 presents our results on image understanding benchmarks spanning fine-grained perception, spatial understanding, and compositional reasoning. Visual Jigsaw achieves substantial improvements across all categories:

Fine-grained Perception. On MMVP Tong et al. (2024), designed to test detailed visual perception, we achieve a +6.00 improvement (54.66 → 60.66). Similarly, MMStar fine-grained tasks show a +6.06 gain (59.75 → 65.81), demonstrating that jigsaw training directly enhances the model's ability to perceive fine visual details.

Spatial Understanding. HR-Bench shows a +3.75 improvement (67.38 → 71.13), while DA-2K achieves a substantial +5.90 gain (54.45 → 60.35). These results indicate that spatial jigsaw training effectively transfers to diverse spatial reasoning tasks.

Compositional Understanding. Even on compositional tasks like Winoground and Sugar-Crepe++, Visual Jigsaw provides modest but consistent improvements (+2.00 and +1.43 respectively), suggesting that enhanced visual perception benefits higher-level reasoning.

RL vs. SFT Comparison. The comparison between Visual Jigsaw (RL) and supervised fine-tuning (SFT) reveals interesting patterns. While SFT shows modest gains on some benchmarks, it significantly hurts performance on others (e.g., LISA-Grounding: 71.89 → 66.59). In contrast, RL training provides consistent improvements across all benchmarks, with particularly strong gains on perception-intensive tasks.

4.3 Video Understanding Results

Table 2: **Evaluation results on video benchmarks.** Video Jigsaw consistently improves over the baseline across all benchmarks and frame settings.

Model	Frames	AoTBench	Vinoground	TOMATO	FAVOR-Bench	TUNA-Bench	VideoMME	TempCompass	TVBench	MotionBench	LVBench	VSI-Bench	Video-TT	CVBench
		vqa	group	test	test	test	wo subs	mc	test	val	test	test	mcq	test
Video-R1	16	45.06	9.40	27.29	49.47	53.00	56.62	70.19	51.80	55.82	34.53	34.34	42.95	47.50
Video-R1	32	47.53	10.20	27.29	49.90	54.26	59.88	71.77	53.54	56.12	38.61	35.11	42.63	48.10
Video-R1	64	48.68	10.60	27.36	50.51	54.33	60.85	72.59	53.43	56.09	38.80	36.61	42.74	48.69
Qwen2.5-VL-7B	16	45.52	12.60	25.87	48.54	53.14	57.44	71.77	49.94	55.56	33.51	32.79	38.39	47.70
Qwen2.5-VL-7B	32	49.48	18.20	26.34	49.34	54.88	60.70	72.59	51.96	56.47	39.19	35.34	41.57	49.60
Qwen2.5-VL-7B	64	52.41	21.80	26.35	50.86	55.79	63.44	72.84	53.74	56.29	40.35	37.74	42.25	51.50
Video Jigsaw	16	51.67	15.20	27.56	49.69	55.10	58.07	73.10	51.33	56.87	36.41	35.39	40.19	49.80
(Gain)		+6.15	+2.60	+1.69	+1.15	+1.96	+0.63	+1.33	+1.39	+1.31	+2.90	+2.60	+1.80	+2.10
Video Jigsaw	32	55.00	21.40	28.03	50.56	56.49	62.37	73.60	53.31	57.99	39.70	38.47	43.27	51.60
(Gain)		+5.52	+3.20	+1.69	+1.22	+1.61	+1.67	+1.01	+1.35	+1.52	+0.51	+3.13	+1.70	+2.00
Video Jigsaw	64	57.64	25.20	28.30	52.27	56.63	64.74	73.60	54.18	57.91	41.83	40.40	44.11	54.50
(Gain)		+5.23	+3.40	+1.95	+1.41	+0.84	+1.30	+0.76	+0.44	+1.62	+1.48	+2.66	+1.86	+3.00

Table 2 shows results across video understanding benchmarks for different frame counts (16, 32, 64). Video Jigsaw demonstrates consistent improvements across temporal reasoning, action understanding, and motion perception tasks:

Temporal Reasoning. AoTBench, which requires understanding action sequences over time, shows substantial improvements: +6.15 for 16 frames, +5.52 for 32 frames, and +5.23 for 64 frames. This demonstrates that temporal jigsaw training enhances temporal understanding capabilities.

Motion and Action Understanding. MotionBench shows consistent gains (+2.90, +0.51, +1.48), while CVBench achieves +2.10, +2.00, and +3.00 improvements respectively. These results indicate that learning to order temporal clips enhances the model's understanding of motion patterns and action sequences.

Frame Count Analysis. Interestingly, improvements tend to be larger for fewer frames, suggesting that the temporal jigsaw task is particularly effective when working with shorter sequences that require precise temporal understanding.

4.4 3D Understanding Results

Table 3: **Evaluation results on 3D benchmarks.** 3D Jigsaw consistently enhances performance on both directly related depth comparison tasks (DA-2K) and broader 3D perception tasks spanning single-view, multi-view, and egocentric video inputs.

Model	SAT-Real	3DSRBench	ViewSpatial	All-Angles	OmniSpatial	VSI-Bench	SPARBench	DA-2K
	test	test	test	test	test	test	tiny	test
Qwen2.5-VL-7B	48.66	57.42	36.52	47.56	42.66	37.74	35.75	54.45
3D Jigsaw	64.00	58.13	38.62	49.06	45.99	40.64	38.31	71.56
(Gain)	+15.34	+0.71	+2.10	+1.50	+3.33	+2.90	+2.56	+17.11

Table 3 presents results on 3D understanding benchmarks. Visual Jigsaw shows particularly strong improvements in 3D spatial reasoning:

Depth-based Tasks. DA-2K, which directly tests depth comparison capabilities, shows a remarkable +17.11 improvement (54.45 $\rightarrow$ 71.56). SAT-Real achieves an equally impressive +15.34 gain (48.66 $\rightarrow$ 64.00). These substantial improvements demonstrate that depth-ordering jigsaw training directly transfers to 3D perception tasks.

Multi-view Understanding. ViewSpatial (+2.10) and All-Angles (+1.50) show consistent improvements, indicating that 3D jigsaw training enhances spatial reasoning across different viewpoints.

General 3D Benchmarks. Even on broader 3D understanding tasks like OmniSpatial (+3.33) and 3DSRBench (+0.71), Visual Jigsaw provides meaningful improvements, suggesting that depth-ordering skills transfer to diverse 3D reasoning scenarios.

4.5 Reasoning Model Analysis

Table 4: Performance of Visual Jigsaw on a reasoning-oriented MLLM (ThinkLite-VL), showing improved visual perception while preserving reasoning ability.

Model	Vision-Centric Avg	MathVista testmini	MathVision testmini	MathVerse testmini	MMMU val	EMMA mini
ThinkLite-VL	59.69	75.20	30.92	50.76	55.11	26.75
+ Image Jigsaw	61.60	75.10	35.20	50.50	54.22	29.00

To evaluate whether Visual Jigsaw complements reasoning-focused models, we apply Image Jigsaw to ThinkLite-VL, a reasoning-oriented MLLM. Table 4 shows that Visual Jigsaw enhances visual perception (Vision-Centric Avg: 59.69 $\rightarrow$ 61.60) while preserving reasoning capabilities on mathematical and scientific tasks.

This result is crucial as it demonstrates that Visual Jigsaw is complementary to reasoning-focused post-training approaches rather than competing with them. The orthogonal nature of perception enhancement allows it to be combined with existing reasoning methods.

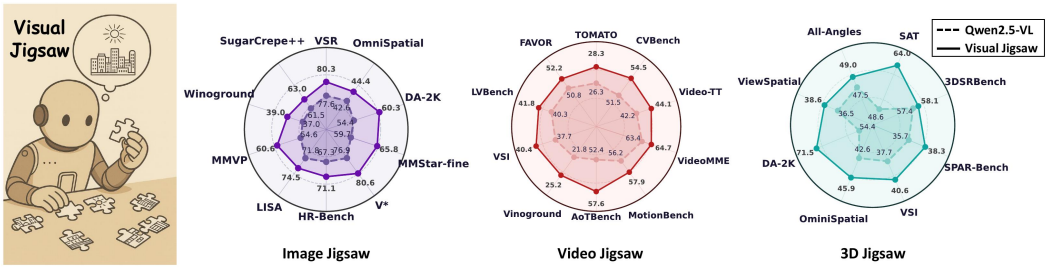

Figure 3: **Visual Jigsaw performance overview.** Our method shows consistent improvements across image, video, and 3D understanding tasks, demonstrating broad applicability for enhancing visual perception in MLLMs.

4.6 Ablation Studies

Jigsaw Difficulty. Figure 2 shows that increasing jigsaw difficulty (more patches for images, more clips for videos) generally leads to better performance. This suggests that harder puzzles provide richer training signals, though there are diminishing returns beyond certain complexity levels.

Reward Function Components. Ablations on the reward function (not shown due to space constraints) reveal that all three components—exact match, partial credit, and format constraints—are necessary for optimal performance. Removing partial credit leads to slower convergence, while removing format constraints causes reward hacking through invalid outputs.

Training Steps and Learning Rate. Visual Jigsaw converges relatively quickly compared to reasoning-focused RL, typically reaching plateau performance within 2-3K training steps. This efficiency stems from the well-defined, verifiable nature of the jigsaw task.

4.7 Analysis and Discussion

Our results demonstrate several key insights:

Modality-Specific Transfer. Each jigsaw variant (spatial, temporal, depth) shows strongest improvements on corresponding benchmark categories, indicating that the learning is appropriately targeted rather than providing generic improvements.

RL Superiority. Across all experiments, RL training consistently outperforms supervised fine-tuning, particularly on perception-intensive tasks. We hypothesize that the exploration inherent in RL allows the model to discover more robust visual understanding strategies.

Complementarity with Reasoning. The success with ThinkLite-VL demonstrates that Visual Jigsaw addresses a distinct aspect of multimodal understanding that is orthogonal to reasoning capabilities, making it a valuable complement to existing post-training methods.

5 Conclusion

We introduced Visual Jigsaw, a novel self-supervised post-training framework that enhances visual perception in multimodal large language models through verifiable permutation prediction tasks. By recasting spatial, temporal, and depth understanding as ordering problems solvable through text generation, Visual Jigsaw provides a unified approach to strengthening intrinsic visual capabilities without requiring architectural modifications or additional generative components.

Our comprehensive evaluation demonstrates substantial improvements across diverse visual understanding benchmarks: +6.00 on MMVP fine-grained perception, +5.23 on AoTBench temporal reasoning, and +17.11 on DA-2K depth understanding. Crucially, these gains come without degrading reasoning capabilities, as evidenced by our experiments with reasoning-oriented models.

The consistent superiority of RL training over supervised fine-tuning highlights the importance of exploration and graded rewards in learning complex visual relationships.

The success of Visual Jigsaw points to a broader principle: effective multimodal understanding requires targeted optimization of both reasoning and perception capabilities. While recent RL approaches have focused primarily on enhancing reasoning through chain-of-thought processes, our work demonstrates that perception-specific training can provide orthogonal and complementary benefits.

Limitations. Visual Jigsaw requires careful tuning of puzzle difficulty and reward function components. The method also focuses specifically on ordering relationships, which, while broadly applicable, may not capture all aspects of visual understanding. Additionally, our evaluation is primarily conducted on a single model family, though the plug-and-play nature of our approach suggests broad applicability.

Future Work. Several directions merit exploration: (1) Extending Visual Jigsaw to other visual relationships beyond ordering, such as part-whole relationships or causal dependencies. (2) Investigating adaptive difficulty scheduling that adjusts puzzle complexity during training. (3) Exploring multimodal jigsaw tasks that require cross-modal reasoning between vision and language. (4) Developing theoretical frameworks to understand why jigsaw tasks are particularly effective for visual perception learning.

Visual Jigsaw represents a significant step toward more comprehensive multimodal understanding, demonstrating that perception and reasoning can be jointly optimized through complementary post-training approaches. As multimodal AI systems become increasingly sophisticated, such targeted enhancement methods will be crucial for achieving human-level visual understanding capabilities.

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